

The moment decomposition for time-varying Johnson–Mehl–Avrami–Kolmogorov kinetics

Documentation for the JMAK per-voxel machinery in THAMES
namespace `jmak` in `JMAKGrowth.h/.cc`

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1 Purpose

This note explains how the four moment accumulators M_0, M_1, M_2, M_3 used in the THAMES JMAK code (`jmak::GlobalMoments`) reduce the doubly-nested time integral of extended volume to a per-cycle constant-time update, when both the nucleation rate $J(t)$ and the linear growth velocity $G(t)$ depend on time. The classical constant- J , constant- G result is assumed familiar; we build on it directly and derive the moment identity used in the implementation.

The technique complements two treatments the reader may already know: Scherer and Bellmann[?] handle time-varying driving force via an *effective time* variable $\xi(t)$; Bullard, Scherer, and Thomas[?] handle it in HydratiCA by discretizing rate laws numerically at each time step. The moment decomposition is a third route: it is algebraic (no change of variable, no numerical quadrature over history) and matches the structure of the JMAK integrand exactly for integer $n = 4$ three-dimensional continuous nucleation, which is the case implemented in the current THAMES code.

2 Classical JMAK, briefly

For constant J (events per unit volume per unit time) and constant G (linear growth velocity of the nucleus–matrix interface), the Johnson–Mehl–Avrami–Kolmogorov result for the transformed volume fraction in a bulk sample is

$$X(t) = 1 - \exp(-Kt^n), \quad (1)$$

where $K = (\pi/3) J G^3$ and $n = 4$ for three-dimensional isotropic growth with continuous (site-nonsaturating) nucleation; $n = 3$ if all nuclei are present at $t = 0$ (site-saturated). Equation (??) follows from the observation that the naive “extended” volume fraction $Y_e(t)$, ignoring impingement between growing particles, is

$$Y_e(t) = \frac{4\pi}{3} \int_0^t J [G(t-\tau)]^3 d\tau = \frac{\pi}{3} J G^3 t^4, \quad (2)$$

and the transformation from extended to actual is

$$X(t) = 1 - \exp[-Y_e(t)], \quad (3)$$

Kolmogorov’s Poisson-impingement correction. The integrand of (??) is (*events per unit volume per unit time*) $\times$ (*volume of a spherical nucleus that has grown for $t - \tau$*), integrated over the birth time τ .

3 Time-varying J and G

When the driving force changes with time, both J and G become functions of t (through the current saturation index or supersaturation, which in cement systems is governed by the coupled evolution of the dissolving reactants and the solid product). The nucleus born at τ then has, at time $t > \tau$, a radius

$$r(\tau, t) = \int_{\tau}^t G(s) ds, \quad (4)$$

which is not simply $G(t - \tau)$ because G varies over the growth interval. The extended volume in a fixed system volume becomes the doubly-nested integral

$$Y_e(t) = \frac{4\pi}{3} \int_0^t J(\tau) \underbrace{\left[\int_{\tau}^t G(s) ds \right]}_{r(\tau, t)}^3 d\tau. \quad (5)$$

Equation (??) is what a naive discretization would carry: at each cycle t_k we would need to evaluate the outer integral by iterating over every past cycle, and for each we would evaluate the inner integral over the interval $[\tau, t_k]$. The cost per cycle is proportional to the number of past cycles, so total work over N cycles is $O(N^2)$. For $N \sim 10^4$ cycles this is 10^8 operations per phase per cycle; expensive but not prohibitive. The moment identity below reduces it to $O(1)$ per cycle per phase without approximation, which matters when the code has multiple phases and the run is long.

4 Change of accumulator

Introduce the cumulative growth arc

$$r_{\text{acc}}(t) = \int_0^t G(s) ds. \quad (6)$$

This is a length, not a rate; it is the radius that a hypothetical nucleus born at $t = 0$ would have reached by time t if G had continued at its actual (possibly time-varying) values. For a nucleus actually born at $\tau > 0$, its radius at time t follows immediately from (??):

$$r(\tau, t) = r_{\text{acc}}(t) - r_{\text{acc}}(\tau). \quad (7)$$

The *difference* of the same accumulator, evaluated at the current time and at the birth time. This is the key.

Substitute (??) into (??):

$$Y_e(t) = \frac{4\pi}{3} \int_0^t J(\tau) [r_{\text{acc}}(t) - r_{\text{acc}}(\tau)]^3 d\tau. \quad (8)$$

The cube in the integrand can now be expanded by the binomial theorem. The term $r_{\text{acc}}(t)$ is independent of the integration variable τ , so it can be pulled out of the integral, exposing the τ -dependence entirely in powers of $r_{\text{acc}}(\tau)$.

5 Moment decomposition

Expanding the cube,

$$\begin{aligned} [r_{\text{acc}}(t) - r_{\text{acc}}(\tau)]^3 &= r_{\text{acc}}(t)^3 - 3 r_{\text{acc}}(t)^2 r_{\text{acc}}(\tau) \\ &\quad + 3 r_{\text{acc}}(t) r_{\text{acc}}(\tau)^2 - r_{\text{acc}}(\tau)^3. \end{aligned} \quad (9)$$

Substituting into (??) and separating t -dependent factors from τ -dependent factors,

$$\begin{aligned} Y_e(t) = \frac{4\pi}{3} &\left[r_{\text{acc}}(t)^3 \int_0^t J(\tau) d\tau - 3 r_{\text{acc}}(t)^2 \int_0^t J(\tau) r_{\text{acc}}(\tau) d\tau \right. \\ &\quad \left. + 3 r_{\text{acc}}(t) \int_0^t J(\tau) r_{\text{acc}}(\tau)^2 d\tau - \int_0^t J(\tau) r_{\text{acc}}(\tau)^3 d\tau \right]. \end{aligned} \quad (10)$$

Define the four *moments* of the accumulated growth radius:

$$M_k(t) = \int_0^t J(\tau) r_{\text{acc}}(\tau)^k d\tau, \quad k \in \{0, 1, 2, 3\}. \quad (11)$$

$M_k(t)$ is the number-density-weighted k -th moment of the growth radius at time t : how many nuclei have been produced up to t , weighted by the k -th power of the accumulated growth radius they have seen since $t = 0$.

With this definition, (??) becomes

$$Y_e(t) = \frac{4\pi}{3} [r_{\text{acc}}(t)^3 M_0(t) - 3 r_{\text{acc}}(t)^2 M_1(t) + 3 r_{\text{acc}}(t) M_2(t) - M_3(t)].$$

(12)

This is the identity implemented in `jmak::extendedVolumePerVoxel` (with `alpha` for the morphology coefficient in place of $4\pi/3$). Each moment $M_k(t)$ satisfies the trivial update rule

$$M_k(t + \Delta t) = M_k(t) + J(t) r_{\text{acc}}(t)^k \Delta t, \quad (13)$$

at first order in Δt , which is one addition and k multiplications per cycle. Similarly $r_{\text{acc}}(t + \Delta t) = r_{\text{acc}}(t) + G(t) \Delta t$. Once a phase's r_{acc} and $M_0, \dots, M_3$ are updated for the current cycle, the extended volume is one arithmetic evaluation of (??): five scalars in, one scalar out, no history lookup.

For a single sample of volume V , the transformed fraction is the usual $X(t) = 1 - \exp[-Y_e(t)/V]$; in the THAMES per-voxel setting, V is the volume of a single lattice voxel and each seeded voxel carries its own M_k snapshot taken at seed time (its “generation” in the code) so that a voxel's own Y_e starts from zero when it seeds, not from $t = 0$. The generation snapshot is subtracted: $Y_e^{\text{gen}}(t) = Y_e(t)$; using $M_k(t) - M_k(t_{\text{seed}})$.

6 Per-cycle numerical update

Only five scalars are stored per phase, plus a small snapshot per generation. The per-cycle work is:

- Evaluate $J(t)$ from the CNT rate law at current $S(t)$ (`cnt::nucleationRate`).
- Evaluate $G(t)$ from the phase's surface-normal rate law times the molar volume: $G = r_{\text{surface}}(S) v_{\text{molar}}$ (`getGrowthVelocity` in each concrete `KineticModel` subclass).

- Advance r_{acc} by $G \Delta t$.
- Advance each M_k by $J r_{\text{acc}}^k \Delta t$.
- For each generation, evaluate (??) using the difference between the current M_k and the generation's snapshot; convert to X_{gen} via $-\text{expm1}(-Y_e/V)$.

None of these steps looks up the past values of J or G . The full growth history is encoded in r_{acc} and the four moments.

7 Connection with Scherer's effective-time formulation

Scherer and Bellmann[?] handle a time-varying driving force by defining an effective time

$$\xi(t) = \int_0^t (\beta_{\text{Tot}}(t')^{1/m} - 1)^n dt', \quad (14)$$

where β_{Tot} is the total-saturation ratio. In their treatment, the linear dimension of each particle is $x_i = G_{0i} \xi$: the entire time evolution is absorbed into a single monotone variable ξ , and the classical JMAK expressions for constant growth rate apply in the ξ coordinate. The two devices — Scherer's effective time and this note's moment decomposition — are addressing the same computational problem (algebraic simplification of a double integral with a time-varying kernel) via different mathematical routes. Scherer's change of variable is exact when the shape factor g (their Eq. 8) and the ratio G_{0i}/G_{0j} are constants, i.e., when the driving-force time dependence is separable from the direction dependence. The moment decomposition presented here is exact when the growth kernel $r(\tau, t) = r_{\text{acc}}(t) - r_{\text{acc}}(\tau)$ separates into t - and τ -dependent pieces after binomial expansion, which is possible for integer positive powers of the kernel. The two coincide in the special case where the whole time-dependence of the rates is captured by a single ξ .

8 Extensions and limitations

- **Non-integer exponent n .** The moment identity (??) requires $[r_{\text{acc}}(t) - r_{\text{acc}}(\tau)]^m$ to be expandable as a finite sum of products of t -only and τ -only factors. This is possible for integer m (via the binomial theorem). For non-integer m — physically defensible for non-spherical or fractal morphologies where $n \in [2.5, 4]$ — the identity does not close, and per-cohort numerical integration of the growth kernel is required. The THAMES code currently implements only $m = 3$ (equivalently $n = 4$); non-integer n is planned as an extension.
- **Site-saturated nucleation.** If all nuclei are placed instantaneously at $t = t_c$ and no further nucleation occurs, $J(t) = N_0 \delta(t - t_c)$ and $M_k(t) = N_0 r_{\text{acc}}(t_c)^k$ for $t \geq t_c$. The formulation still applies, and the classical result $Y_e = (4\pi/3) N_0 r(t_c, t)^3$ is recovered.
- **Non-spherical shapes and heterogeneous nucleation.** The morphology coefficient α replaces $4\pi/3$ for non-spherical growth (e.g., $\alpha \approx \pi g/2$ for blade-like C-S-H per Scherer's shape factor). For heterogeneous nucleation on a scarce substrate the effective J must be further weighted by the fraction of available substrate; the moment update still applies to the effective J , but the substrate-availability multiplier is a separate factor whose time dependence must be tracked. In THAMES, the CNT rate law `cnt::nucleationRate` already carries a θ parameter for that (Turnbull–Vonnegut $f(\theta)$).

- **Impingement between voxels.** JMAK’s Poisson-impingement correction $X = 1 - \exp(-Y_e/V)$ operates *within a voxel*. Impingement between neighboring lattice voxels is handled at the placement stage (`Lattice::nucleatePhaseRnd`): once a voxel is fully transformed ($X_{\text{gen}} \approx 1$), its Portlandite contribution is committed as a whole voxel of the phase and no further JMAK evolution applies to it. Growth into neighboring electrolyte is handled by the standard THAMES rate-law/ lattice-face mechanics.

References

- [1] G. W. Scherer, F. Bellmann, *Kinetic analysis of C-S-H growth on calcite*, Cement and Concrete Research **103**, 226–235 (2018).
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- [4] I. Avramov, J. Šesták, Generalized kinetics of non-isothermal transformation phenomena, *Journal of Thermal Analysis and Calorimetry* **79**, 595–602 (2005).